

# Exercise 1. Red Wine Quality Dataset

## Red Wine Quality Dataset

### 1.a Reading the Dataset

Read the dataset into an R object. Make sure the column headers are correct, and show the first couple rows.

```
library(tidyverse)
library(kableExtra)

red_wine_data <- read.csv(
  "data/wine+quality/winequality-red.csv",
  sep = ";"
)

# Display the first six observations of the first six variables
table_1 <- knitr::kable(
  head(red_wine_data[, 1:6]),
  caption = "First six observations of the red wine quality dataset."
) %>%
  kableExtra::kable_styling(latex_options = "hold_position")

# Display the first six observations of the last six variables
table_2 <- knitr::kable(
  head(red_wine_data[, 7:12]),
  caption = "First six observations of the red wine quality dataset."
) %>%
  kableExtra::kable_styling(latex_options = "hold_position")

table_1
```

Table 1: First six observations of the red wine quality dataset.

| fixed.acidity | volatile.acidity | citric.acid | residual.sugar | chlorides | free.sulfur.dioxide |
|---------------|------------------|-------------|----------------|-----------|---------------------|
| 7.4           | 0.70             | 0.00        | 1.9            | 0.076     | 11                  |
| 7.8           | 0.88             | 0.00        | 2.6            | 0.098     | 25                  |
| 7.8           | 0.76             | 0.04        | 2.3            | 0.092     | 15                  |
| 11.2          | 0.28             | 0.56        | 1.9            | 0.075     | 17                  |
| 7.4           | 0.70             | 0.00        | 1.9            | 0.076     | 11                  |
| 7.4           | 0.66             | 0.00        | 1.8            | 0.075     | 13                  |

table\_2

Table 2: First six observations of the red wine quality dataset.

| total.sulfur.dioxide | density | pH   | sulphates | alcohol | quality |
|----------------------|---------|------|-----------|---------|---------|
| 34                   | 0.9978  | 3.51 | 0.56      | 9.4     | 5       |
| 67                   | 0.9968  | 3.20 | 0.68      | 9.8     | 5       |
| 54                   | 0.9970  | 3.26 | 0.65      | 9.8     | 5       |
| 60                   | 0.9980  | 3.16 | 0.58      | 9.8     | 6       |
| 34                   | 0.9978  | 3.51 | 0.56      | 9.4     | 5       |
| 40                   | 0.9978  | 3.51 | 0.56      | 9.4     | 5       |

## 1.b Summary Statistics

```
median_quality <- median(red_wine_data$quality)
mean_alcohol <- mean(red_wine_data$alcohol)
```

**1. Median Quality Score: 6**

**2. Mean Alcohol Level: 10.4229831**

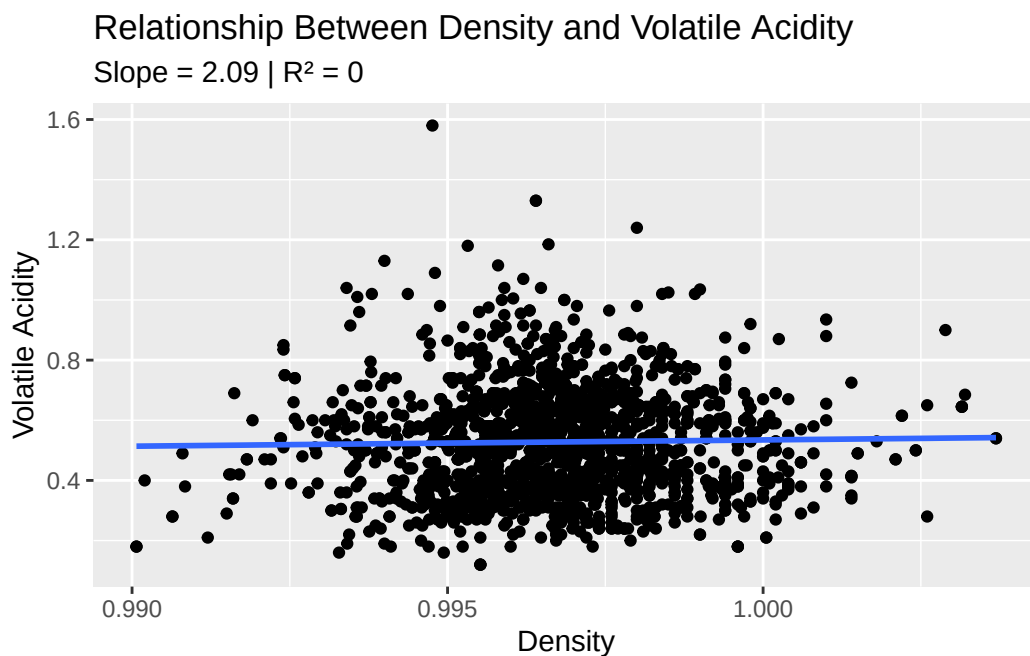
## 1.c Density and Volatile Acidity

Create a scatterplot showing the relationship between density and volatile acidity.

```
# Calculate the linear regression model
model <- lm(volatile.acidity ~ density, data = red_wine_data)

# Extract slope and R-squared
slope <- coef(model)["density"]
r_squared <- summary(model)$r.squared

ggplot(red_wine_data, aes(x = density, y = volatile.acidity)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Relationship Between Density and Volatile Acidity",
    subtitle = paste(
      "Slope =", round(slope, 3),
      "| R² =", round(r_squared, 3)
    ),
    x = "Density",
    y = "Volatile Acidity"
  )
```



The best-fit line has a slope of **2.09**, which indicates a positive relationship between density and volatile acidity. However, With an  $R^2$  value of **0**, the relationship is very weak. There is essentially no relationship between the two variables.

## 1.d Alcohol Level Categories

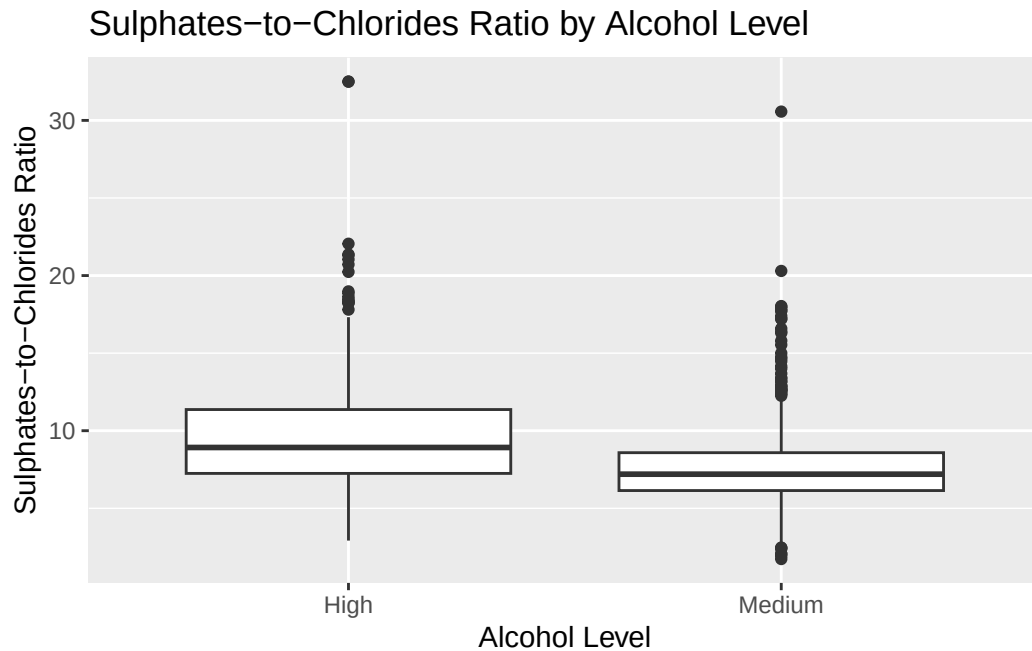
Classify each wine sample according to the following conditions:

- **High:** Alcohol level is greater than 10.5
- **Medium:** Alcohol level is less than or equal to 10.5.

```
# Create the ALevel categorical variable
red_wine_data$ALevel <- ifelse(
  red_wine_data$alcohol > 10.5,
  "High",
  "Medium"
)

# Calculate the sulphates-to-chlorides ratio
red_wine_data$sulphates_to_chlorides <-
  red_wine_data$sulphates / red_wine_data$chlorides

# Create side-by-side boxplots
ggplot(
  red_wine_data,
  aes(x = ALevel, y = sulphates_to_chlorides)
) +
  geom_boxplot() +
  labs(
    title = "Sulphates-to-Chlorides Ratio by Alcohol Level",
    x = "Alcohol Level",
    y = "Sulphates-to-Chlorides Ratio"
  )
```



```
# Count the number of High ALevel samples
high_count <- sum(red_wine_data$ALevel == "High")
```

The **High** group has a higher median sulphates-to-chlorides ratio than the **Medium** group. The **High** group also shows greater variability. Both groups contain several high-level outliers. There are **616** wine samples in the **High** ALevel category.

### 1.e Citric Acid Distribution

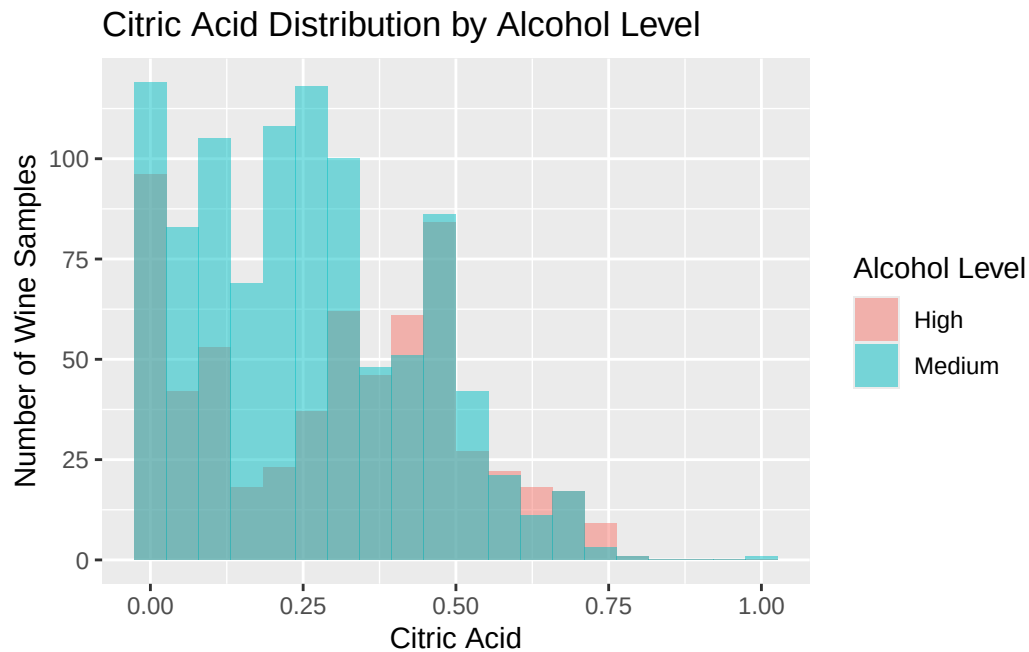
Create a histogram showing the citric acid values for both the **High** and **Medium** groups. If both groups are shown in the same plot, include a legend to distinguish them.

```
ggplot(
  red_wine_data,
  aes(x = citric.acid, fill = ALevel)
) +
  geom_histogram(
    bins = 20,
    alpha = 0.5,
    position = "identity"
  ) +
  labs(
```

```

title = "Citric Acid Distribution by Alcohol Level",
x = "Citric Acid",
y = "Number of Wine Samples",
fill = "Alcohol Level"
)

```



## 1.f Additional Exploration

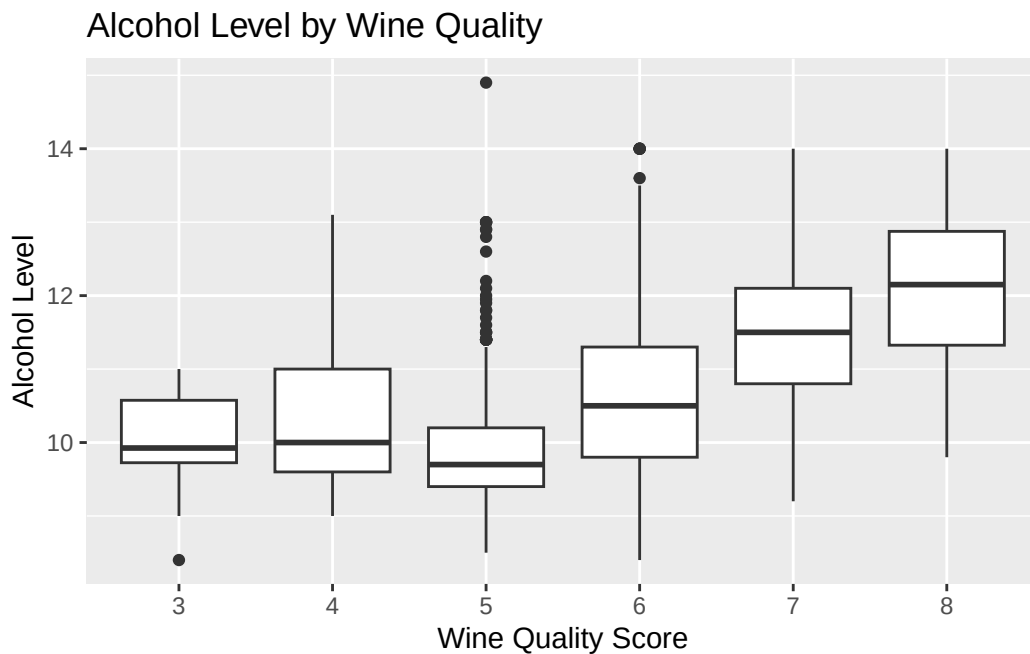
Continue exploring the dataset by creating two additional plots. For each plot:

- State the question or hypothesis.
- Include the code and resulting plot.
- Summarize what you observe.

### Plot 1: Alcohol Level and Wine Quality

**Question:** Does alcohol level vary across different wine qualities?

```
ggplot(red_wine_data, aes(x = factor(quality), y = alcohol)) +
  geom_boxplot() +
  labs(
    title = "Alcohol Level by Wine Quality",
    x = "Wine Quality Score",
    y = "Alcohol Level"
  )
```



Higher-quality wines generally have higher alcohol levels. The median alcohol level increases noticeably for quality scores 6–8 compared with scores 3–5.

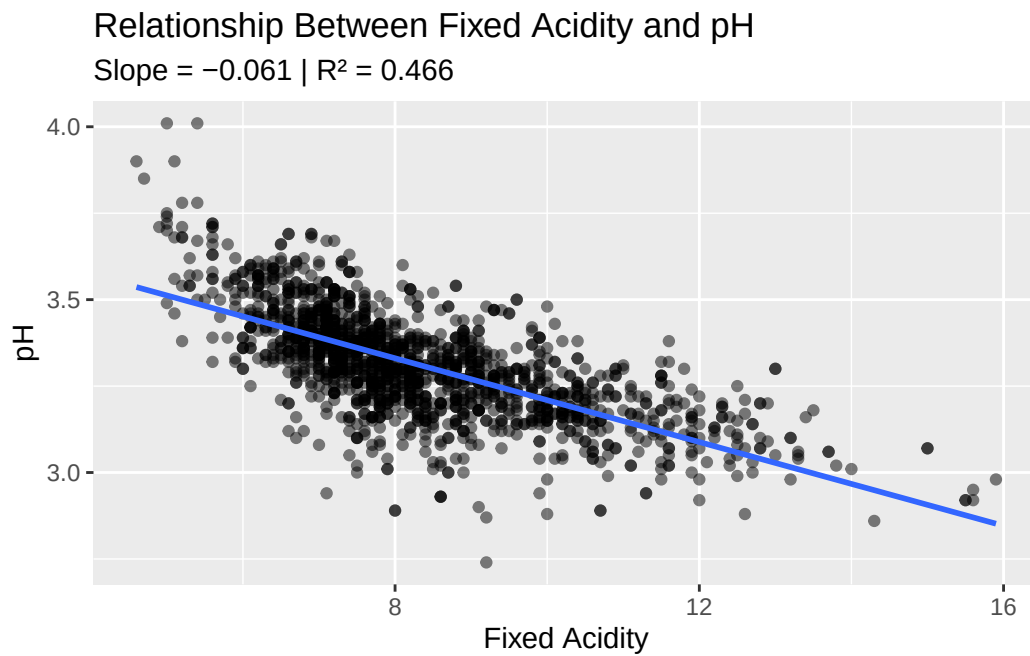
## Plot 2: pH and Fixed Acidity

**Question:** Is there a relationship between fixed acidity and pH?

```
model <- lm(pH ~ fixed.acidity, data = red_wine_data)
slope <- coef(model)["fixed.acidity"]
r_squared <- summary(model)$r.squared

ggplot(red_wine_data, aes(x = fixed.acidity, y = pH)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
```

```
labs(
  title = "Relationship Between Fixed Acidity and pH",
  subtitle = paste(
    "Slope =", round(slope, 3),
    "| R2 =", round(r_squared, 3)
  ),
  x = "Fixed Acidity",
  y = "pH"
)
```



The scatterplot shows a moderate negative linear relationship between fixed acidity and pH. The slope of  $-0.061$  indicates that pH decreases as fixed acidity increases, while the  $R^2$  value of  $0.466$  indicates that fixed acidity explains approximately 46.6% of the variation in pH.